

INTRODUCTION OF COMPANY

1. Company building having three floors like ground floor, 1st floor and second floor.
2. Total 53 devices out of which 46 mobile devices and 07 are fixed.
3. Total No of employees are 21.
4. The workforce includes:
 - 4 Equipment handlers
 - 1 Equipment manager
 - 1 Head chef
 - 10 Chefs
 - 1 Catering manager
 - 3 Office workers
 - CEO (owner)
5. The table below outlines the employees and their associated devices.

	Office Worker	Equipment Handler	Equipment Manager	Transient Staff	Catering Manage	Head Chef	Chefs	CEO	Total devices
Desktop Computer	3		1		1	1		1	7
Laptop Computer							3		3
Tablet		4	1		1	1	3	1	11
Mobile Phone				30	1			1	32

6. VIP Food app description.

The app offers a range of specialized functionalities, including:

 - Equipment management
 - Kitchen management
 - Event management
 - Administration
 - Day-to-day tasks
7. Different employees access to the app as per their roles, such as define below.

Equipment Handlers Role (EquipHandlersRole):

Tailored for equipment handlers, this role provides access to essential equipment management functionalities within the VIP Foods App. It enables efficient tracking and handling of culinary tools and resources.

Equipment Manager Role (EquipManagerRole):

Designed for equipment managers, this role offers advanced capabilities to oversee and manage the full spectrum of equipment functionalities. It ensures optimal utilization and maintenance of culinary resources.

Chefs Role (ChefsRole):

Crafted for chefs, this role grants access to food preparation and kitchen management functionalities. Chefs can leverage the app to streamline culinary processes and enhance collaboration within the kitchen.

Head Chef Role (HeadChefRole):

Specifically designed for head chefs, this role provides elevated access to overall kitchen management functionalities. It empowers head chefs with tools to efficiently coordinate and lead culinary operations.

Catering Manager Role (CateringManagerRole):

Tailored for catering managers, this role facilitates planning and managing catering events. It includes features for event coordination, menu planning, and logistics management to ensure seamless catering services.

Office Workers Role (OfficeWorkersRole):

Customized for office workers, this role provides access to office-related activities and data management functionalities within the VIP Foods App. It enhances communication and collaboration between the culinary and administrative teams.

CEO (Owner) Role (CEORole):

Specifically designed for the CEO or owner, this role grants unrestricted access to all functionalities on the VIP Foods App. It enables strategic oversight, and decision-making, and ensures comprehensive control over the platform.

Temporary Role (TempRole):

Assigned to temporary workers, this role offers limited access to permissions required for the day's tasks. The permissions are automatically disabled after a predefined period, ensuring temporary and secure access tailored to the user's immediate needs.

STAGE 01

In stage 01 we have 06 tasks to do.

Task01 => Understand the Requirements

Task02=> Understand the building structure and planning secure network installation.

Task03=> Access control and Security Policies

Task04=> User roles and provide access

Task05=> Physical Security

Now we discuss on each task in details.

Task 01: Understand the Requirements.

From the introduction it is clear, we must provide cybersecurity to the company and company have 21 employees, some wired and some wireless internet devices, company also have their app. Now we must provide such a security level in which company employees gained access to the app securely as per their role access with physical security of devices and much more security aspect which we discuss in later stages.

Task 02: Understand the building structure and planning secure network installation.

The network architecture for the new building is guided by defense in depth principles to establish a comprehensive layered security approach. Network segmentation should be implemented to isolate critical systems and data, reducing the risk of potential attacks.

- Equipment handlers:
Wired: 10.0.1.0/26
Wireless: 10.0.1.128/26
- Equipment manager:
Wired: 10.0.1.64/26
Wireless: 10.0.1.192/26
- **First floor subnets:**
 - Chefs:
Wired: 10.0.2.0/26
Wireless: 10.0.2.128/26
 - Head chef:
Wired: 10.0.2.64/26
Wireless: 10.0.2.192/26
 - Catering manager:
Wired: 10.0.2.128/26
Wireless: 10.0.2.0/26
- **Second floor subnets:**
 - Office workers:
Wired: 10.0.3.0/26
Wireless: 10.0.3.128/26

- CEO (Owner):
Wired: 10.0.3.64/26
Wireless: 10.0.3.192/26
Production subnet:
- Servers, printers, POS systems:
Wired: 10.1.0.0/24
Wireless: 10.1.0.128/25
Management subnets:
- Routers, switches, firewalls:
Wired: 10.1.1.0/24
Wireless: 10.1.1.128/25
- Guest and transient worker subnet:
Laptops, tablets, smartphones: 10.1.2.0/24
Wireless: 10.1.2.128/25

Task03=> Access control and Security Policies

User access to specific subnets should be facilitated through precisely outlined policies overseen by the on-premises firewall.

install firewalls at the network perimeter to efficiently filter both incoming and outgoing traffic. Firewalls should be configured to analyze and permit legitimate data while proactively blocking potentially harmful content.

deploy intrusion detection/prevention systems (IDS/IPS) at the network perimeter.

Utilize IDS/IPS systems to continuously monitor network activities, ensuring prompt detection of and response to any suspicious behavior or potential intrusion attempts.

Task04=> User roles and provide access

Azure Active Directory (Azure AD) implementation: An Azure AD tenant should be configured and user accounts should be created based on employee roles. It is recommended to use group-based access control with a twist—assign roles directly within the groups. By doing so, group members will automatically inherit the specified role, ensuring a seamless and efficient approach to resource access management according to user groups.

Task05=> Physical Security

This boring me to discuss but let start having dedicated IT Room for server router and switch with restriction of unauthorized access, ID Based Access CCTV Cameras locked bela bela.

STAGE 02

AZURE ACTIVE DIRECTORY SETUP

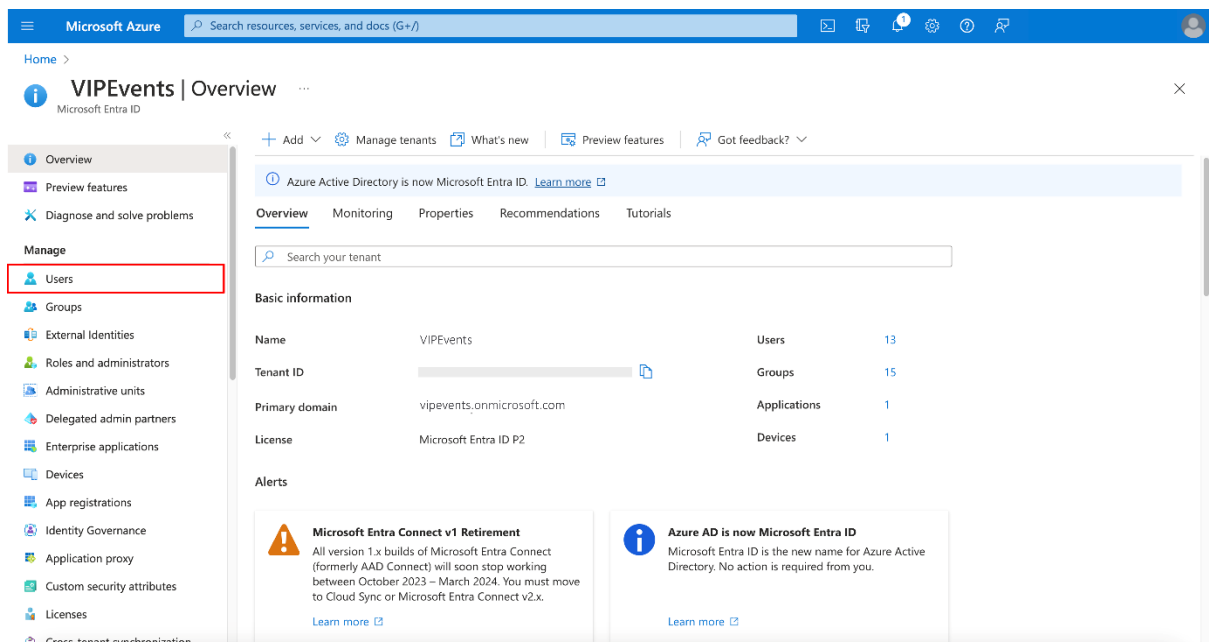
Task 01 => Azure AD Tenant Creation.

An Azure AD tenant should be created for VIP Events to ensure seamless integration with the organization's infrastructure

Task 02 => User Account Configuration

1. User Account Creation

Access the **Microsoft Entra Admin Center** and sign in with privileges equivalent to a User Administrator (at minimum).



2. Navigate to **User Creation**.

3. Scroll down to **Manage**, select **Users** and then **All users**.

14 users found

	Display name ↑	User principal name ↑	User type	On-premises sy...	Identities	Company name
☐	Ravi Patel	equip_handler1@vipeve...	Member	No	vipevents.onmicrosoft.com	VIPEvents
☐	Luis Mendoza	equip_handler2@vipevc...	Member	No	vipevents.onmicrosoft.com	VIPEvents
☐	Marco Rossi	equip_handler3@vipeve...	Member	No	vipevents.onmicrosoft.com	VIPEvents
☐	Benjamin Harris	equip_handler4@vipeve...	Member	No	vipevents.onmicrosoft.com	VIPEvents
☐	Ryan Miller	equip_manager@vipeve...	Member	No	vipevents.onmicrosoft.com	VIPEvents
☐	Fatima Kapoor	chef1@vipevents.ommic...	Member	No	vipevents.onmicrosoft.com	VIPEvents
☐	David Wilson	chef2@vipevents.ommic...	Member	No	vipevents.onmicrosoft.com	VIPEvents
☐	Sofia Costa	chef3@vipevents.ommic...	Member	No	vipevents.onmicrosoft.com	VIPEvents
☐	Emma Davis	chef4@vipevents.ommic...	Member	No	vipevents.onmicrosoft.com	VIPEvents
☐	Chloe White	chef5@vipevents.ommic...	Member	No	vipevents.onmicrosoft.com	VIPEvents
☐	Sarah Lee	chef6@vipevents.ommic...	Member	No	vipevents.onmicrosoft.com	VIPEvents
☐	Mia Robinson	chef7@vipevents.ommic...	Member	No	vipevents.onmicrosoft.com	VIPEvents
☐	Ethan Martinez	chef8@vipevents.ommic...	Member	No	vipevents.onmicrosoft.com	VIPEvents

4. Create a new user by selecting **+New user** at the top of the pane and then select **Create new user**.
5. In the **Basics** tab, add user account details. In the **User principal name** field add the usernames provided in the table below (e.g. “equip_manager@vipevents.onmicrosoft.com”), adjusting the domain as needed.
6. In the **Display name** field, enter the user's name.

Create a new internal user in your organization

Basics Properties Assignments Review + create

Create a new user in your organization. This user will have a user name like alice@contoso.com. [Learn more](#)

Identity

User principal name: equip_manager@vipevents.onmicrosoft.com

Mail nickname: equip_manager

Display name: Ryan Miller

Password: [masked]

Auto-generate password: ☒

Account enabled: ☒

Review + create Previous Next: Properties Give feedback

7. Under the **Properties** tab, add the **Job role** and **Department** provided in the table for each user.

Microsoft Azure Search resources, services, and docs (G+)

Home > VIPEvents > Users > Users >

Create new user

Create a new internal user in your organization

Basics **Properties** Assignments Review + create

Identity

First name

Last name

User type

Authorization info [+ Edit Certificate user IDs](#)

Job Information

Job title

Company name

Department

Employee ID

Employee type

Employee hire date

Review + create < Previous Next: Assignments > [Give feedback](#)

- Under the **Assignments** tab, enter the required details for the user under the **Groups and roles**, **Settings**, and **Job info** sections.

Microsoft Azure Search resources, services, and docs (G+)

Home > VIPEvents > Users > Users >

Create new user

Create a new internal user in your organization

Basics Properties **Assignments** Review + create

Make up to 20 group or role assignments. You can only add a user to a maximum of 1 administrative unit.

[+ Add administrative unit](#) [+ Add group](#) [+ Add role](#)

No assignments to display.

Review + create < Previous Next: Review + create > [Give feedback](#)

- To finalize the user creation, select **Review+Create** and then **Create**.

Microsoft Azure Search resources, services, and docs (G+)

Home > VIPEvents > Users > Users

Create new user

Create a new internal user in your organization

Basics Properties Assignments **Review + create**

Basics

User principal name equip_manager@SamsScoops.onmicrosoft.com

Display name Ryan Miller

Mail nickname equip_manager

Password

Account enabled Yes

Properties

First name Ryan

Last name Miller

User type Member

Job title Equipment manager

Company name VIPEvents

Department Operations

Assignments

Create < Previous Next > Give feedback

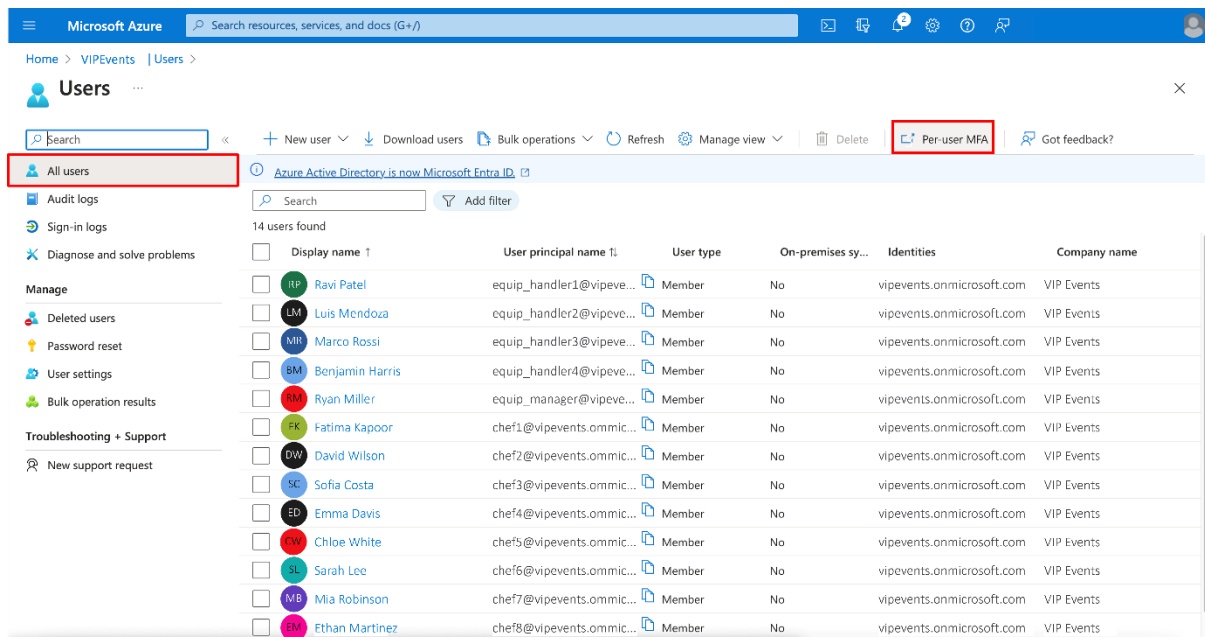
The table below provides user details for VIP Events employees.

Note: Usernames used here were randomly generated, you can use any user names.

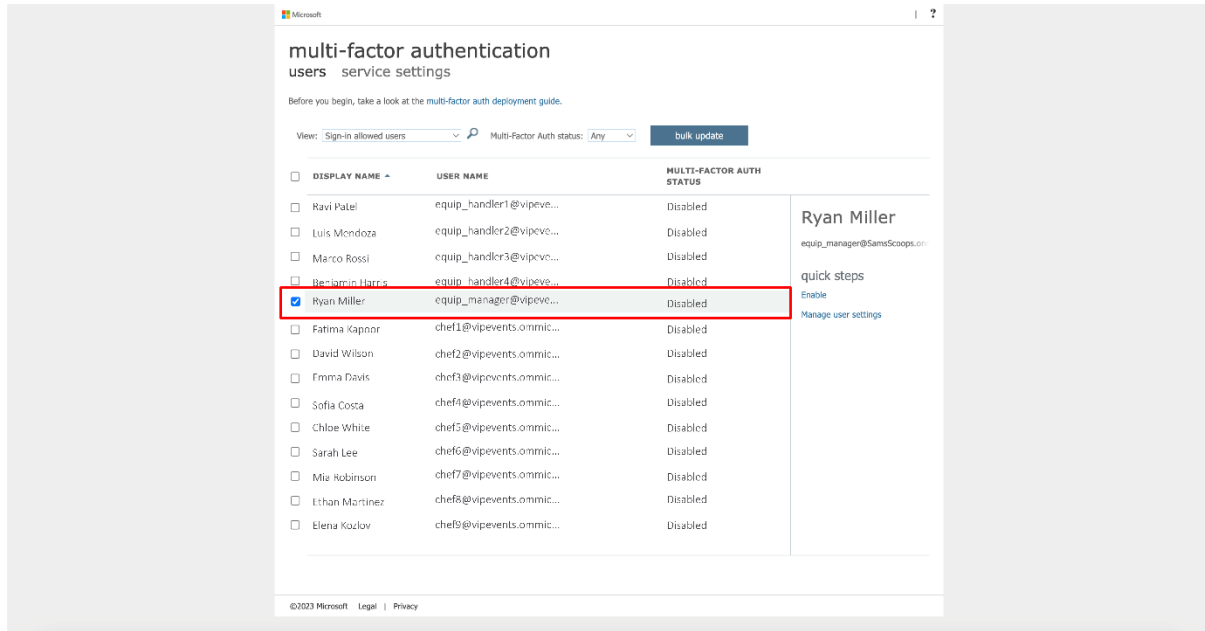
Username	Display name	Job role	Department
equip_handler1	Ravi Patel	Equipment handler	Operations
equip_handler2	Luis Mendoza	Equipment handler	Operations

Security should be enhanced by enabling multi-factor authentication for all user accounts. To do this follow the steps outlined below:

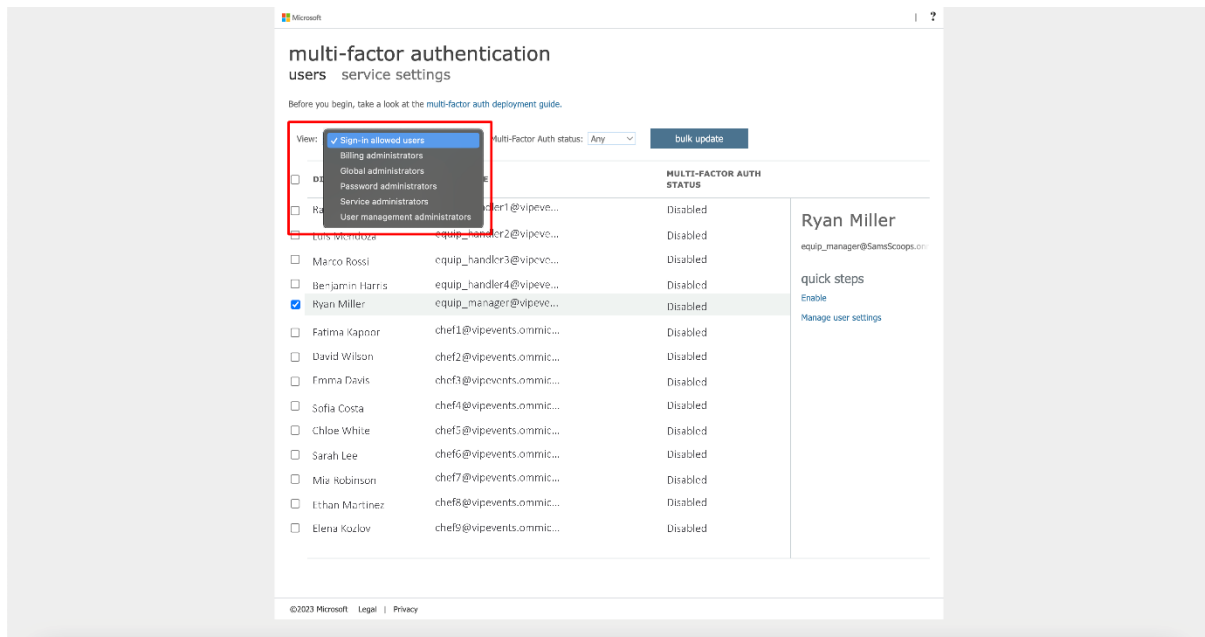
1. Navigate to the Microsoft Entra **multi-factor authentication** users page.



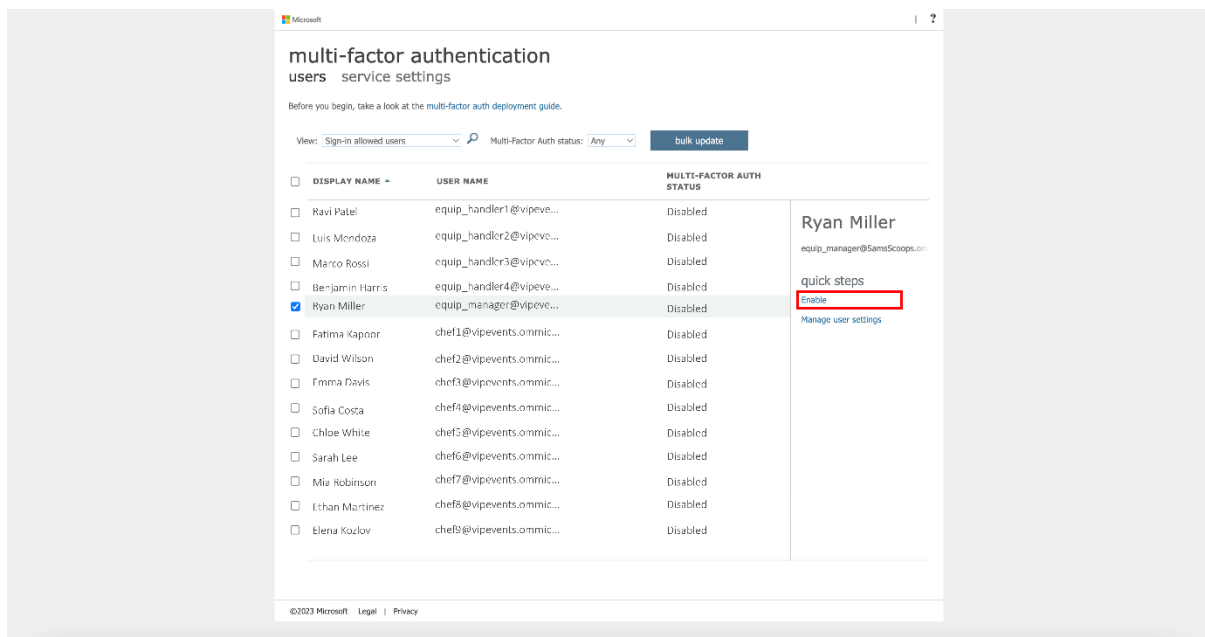
2. To check if MFA is enabled for a user, locate and select the user as demonstrated in the screenshot below.



3. Adjust the view to **User** if needed.



4. Select the user(s) by checking the box next to their names.
5. To enable multi-factor authentication, select **Enable** on the panel on the right of the screen, under **quick steps**.



6. Confirm your selection in the pop-up window that appears by selecting **enable multi-factor auth**.



Task03 => **Group-based access control implementation**

To implement a group-based access control system, users should be categorized into distinct groups based on their roles. The following are the identified user groups along with their respective members:

- Equipment handler group:

Members: equip_handler1, equip_handler2, equip_handler3, equip_handler4

- Equipment manager group:

Members: equip_manager

- Chefs group:

Members: chef1, chef2, chef3, chef4, chef5, chef6, chef7, chef8, chef9, chef10

- Head chef group:

Members: head_chef

- Catering manager group:

Members: catering_manager

- Office workers group:

Members: office_worker1, office_worker2, office_worker3

- CEO (Owner) group:

Members: ceo_owner

- Worker - External:

Members: user1_ext

Dedicated security groups should be established within Azure AD that align with the identified user roles. To do this, follow these steps:

1. Access the **Microsoft Entra Admin Center** as a Global Admin.
2. Navigate to **Group Creation**, scroll down to Identity, select **Groups** and then **All groups**.
3. To initiate a new group, select **New group**.
4. Configure the group details by adding the required group type, name, and description.
5. Enable role assignment for the group: Set **Microsoft Entra roles can be assigned to the group** to **Yes**.

Microsoft Azure

Home > VIVEvents > Groups > Groups | All groups >

New Group

Got feedback?

Group type * ⓘ
Security

Group name * ⓘ
Enter the name of the group

Group description ⓘ
Enter a description for the group

Microsoft Entra roles can be assigned to the group ⓘ
Yes No

Membership type ⓘ
Assigned

Owners
No owners selected

Members
No members selected

Roles
No roles selected

Create

6. Select members and owners for the group.
7. Assign roles to the group.
8. To confirm the group creation, select **Create**.
9. Acknowledge the irreversible nature of the role-assignable group setting when prompted.
10. Finalize the group creation by selecting **Yes**.

Microsoft Azure

Search resources, services, and docs (G+)

[Home](#) > [VIPEvents](#) | [Groups](#) > [Groups](#) | [All groups](#) >

New Group

 Got feedback?

Creating a group to which Microsoft Entra roles can be assigned is a setting that cannot be changed later. Are you sure you want to add this capability? [Learn More.](#)

Equipment Handlers Group

Group description ⓘ

Equipment Handlers Role can be assigned to this group 

Microsoft Entra roles can be assigned to the group ⓘ

☒ Yes ☐ No

Membership type ⓘ

Assigned 

To map user groups to Azure roles, you should integrate individual user accounts into the access control framework by adding them to their corresponding security groups. This ensures a smooth and organized implementation of the group-based access control system.

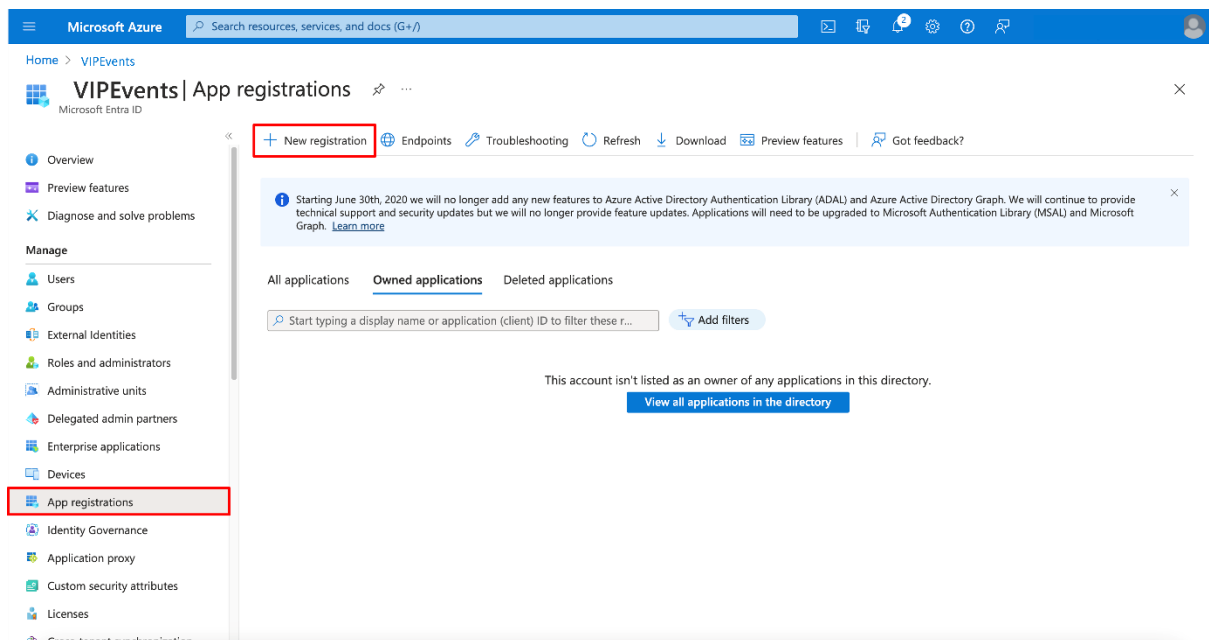
Stage 03 (Define Roles and Access)

This section describes the proposed Azure role and access configuration. These roles, tailored to the unique needs of VIP Events, are strategically assigned based on the user groups established earlier, creating a precise and secure access control mechanism.

VIP Foods application integration (Task 1)

The first step that needs to be performed to allow for role-based access control is to create an application registration for the VIP Food app. The key steps involved are as follows:

1. Sign in to the **Microsoft Entra admin center**
2. Navigate to **Manage**, select **Applications** then **App registrations** and finally select **New registration**.



3. Enter a display name for the application, specify the sign-in audience, register the application, and note the Application ID (client).
4. Add a redirect URI for the VIP Food app.
5. And select **Register**.

Microsoft Azure

Search resources, services, and docs (G+/)

Home > VPEvents > App registrations >

Register an application

* Name

The user-facing display name for this application (this can be changed later).

VIP Food App

Supported account types

Who can use this application or access this API?

☒ Accounts in this organizational directory only (SamsScoops only - Single tenant)

☐ Accounts in any organizational directory (Any Microsoft Entra ID tenant - Multitenant)

☐ Accounts in any organizational directory (Any Microsoft Entra ID tenant - Multitenant) and personal Microsoft accounts (e.g. Skype, Xbox)

☐ Personal Microsoft accounts only

[Help me choose...](#)

Redirect URI (optional)

We'll return the authentication response to this URI after successfully authenticating the user. Providing this now is optional and it can be changed later, but a value is required for most authentication scenarios.

Webhttps://vipfoodapp.com/auth/callback

Register an app you're working on here. Integrate gallery apps and other apps from outside your organization by adding from Enterprise applications.

By proceeding, you agree to the Microsoft Platform Policies

Register

Microsoft Azure

Search resources, services, and docs (G+/)

Home > VPEvents > App registrations >

VIP Food App

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Token configuration

API permissions

Expose an API

App roles

Owners

Roles and administrators

Manifest

Support + Troubleshooting

Troubleshooting

New support request

Essentials

Display name : VIP Food App

Application (client) ID :

Object ID :

Directory (tenant) ID :

Supported account types : My organization only

Client credentials : Add a certificate or secret

Redirect URIs : 1 web, 0 spa, 0 public client

Application ID URI : Add an Application ID URI

Managed application in L... : VIP Food App

Welcome to the new and improved App registrations. Looking to learn how it's changed from App registrations (Legacy)? Learn more

Starting June 30th, 2020 we will no longer add any new features to Azure Active Directory Authentication Library (ADAL) and Azure Active Directory Graph. We will continue to provide technical support and security updates but we will no longer provide feature updates. Applications will need to be upgraded to Microsoft Authentication Library (MSAL) and Microsoft Graph. Learn more

Get StartedDocumentation

Build your application with the Microsoft identity platform

The Microsoft identity platform is an authentication service, open-source libraries, and application management tools. You can create modern, standards-based authentication solutions, access and protect APIs, and add sign-in for your users and customers. Learn more

6. Add a client secret.

Microsoft Azure

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Home > VPEvents > App registrations >

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Application (client) ID :

Object ID :

Directory (tenant) ID :

Supported account types : My organization only

Client credentials : Add a certificate or secret

Redirect URIs : 1 web, 0 spa, 0 public client

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7. Select + New client secret.

Microsoft Azure | Search resources, services, and docs (G+/J)

Home > VPEvents > App registrations > VIP Food App

VIP Food App | Certificates & secrets

Search | Got feedback?

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 - Branding & properties
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 - Certificates & secrets
 - Token configuration
 - API permissions
 - Expose an API
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 - Owners
 - Roles and administrators
 - Manifest
- Support + Troubleshooting
 - Troubleshooting
 - New support request

Credentials enable confidential applications to identify themselves to the authentication service when receiving tokens at a web addressable location (using an HTTPS scheme). For a higher level of assurance, we recommend using a certificate (instead of a client secret) as a credential.

Application registration certificates, secrets and federated credentials can be found in the tabs below.

Certificates (0) **Client secrets (0)** Federated credentials (0)

A secret string that the application uses to prove its identity when requesting a token. Also can be referred to as application password.

+ New client secret

Description	Expires	Value	Secret ID
No client secrets have been created for this application.			

Microsoft Azure | Search resources, services, and docs (G+/J)

Home > VPEvents > App registrations > VIP Food App

VIP Food App | Certificates & secrets

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Application registration certificates, secrets and federated credentials can be found in the tabs below.

Certificates (0) **Client secrets (0)** Federated credentials (0)

A secret string that the application uses to prove its identity when requesting a token. Also can be referred to as application password.

+ New client secret

Description	Expires	Value
No client secrets have been created for this application.		

Add a client secret

Description:

Expires: Recommended: 180 days (6 months)

Add Cancel

Microsoft Azure | Search resources, services, and docs (G+/J)

Home > VPEvents > App registrations > VIP Food App

VIP Food App | Certificates & secrets

Search | Got feedback?

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Credentials enable confidential applications to identify themselves to the authentication service when receiving tokens at a web addressable location (using an HTTPS scheme). For a higher level of assurance, we recommend using a certificate (instead of a client secret) as a credential.

Application registration certificates, secrets and federated credentials can be found in the tabs below.

Certificates (0) **Client secrets (1)** Federated credentials (0)

A secret string that the application uses to prove its identity when requesting a token. Also can be referred to as application password.

+ New client secret

Description	Expires	Value	Secret ID
VIP Food App	5/28/2024		

1. Document these details.

To facilitate clarity and future reference, the following details should be documented.

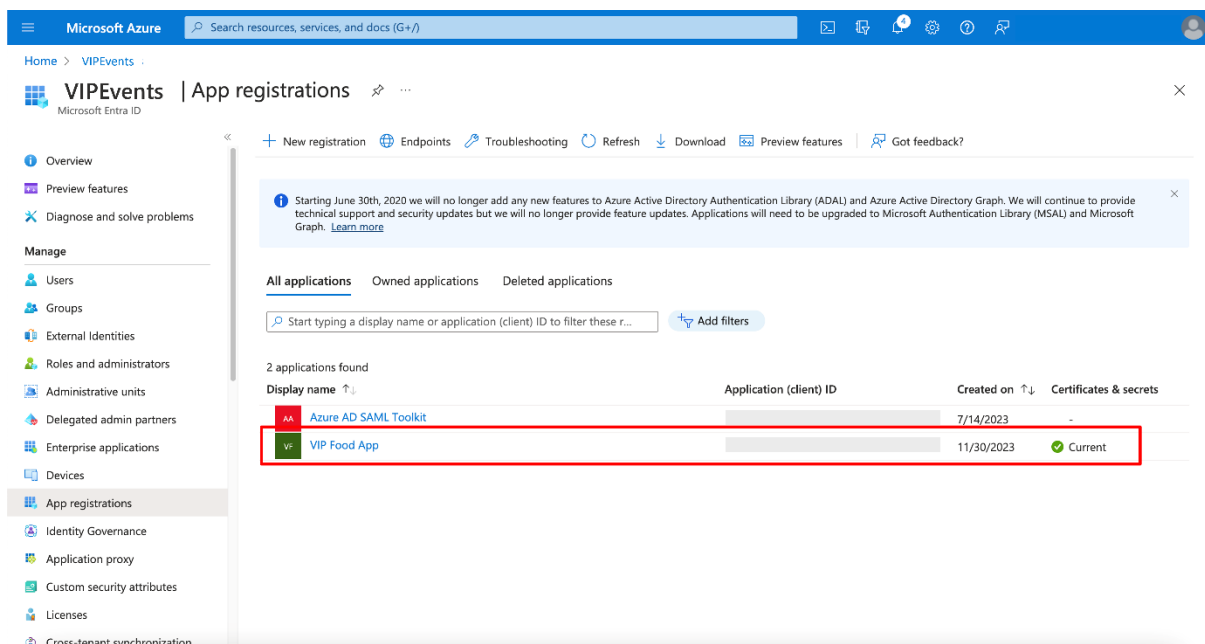
Registered app details:

- Name: VIP Food App
- Client ID: 17546bc3f-f5d3-451a-a45c-6a831a87a5d3
- Tenant ID: 98ce5354-3b6b-9735-b80f-4826b0ccb3a9
- Client secret
 - Description: VIP Food App
 - Expires: 5/28/2024
 - Value: Elb8Q~UmtQcbSXgQ_2dHnClaNOWOTMrP_.P8da69
 - Secret ID: 79e9994f-0197-4607-ad6d-b4fc4342baed
- Redirect URI: Web: `https://{URL}/auth/callback`Stage 4: AAD Connections

Custom Azure Roles (Task 2)

To align roles with user groups and facilitate controlled access, custom Azure roles are needed. The process of creating custom Azure roles involves the following steps:

1. Sign in to the **Microsoft Entra admin center** as a Global Administrator.
2. Navigate to **Identity**, select **Applications**, and then **App registrations**. Select the relevant application here.



3. Under **Manage**, choose **App roles** and then **Create app role**.
4. Configure role settings and parameters in the **Create app role** pane.

Microsoft Azure | Search resources, services, and docs (G+)

Home > VIPEvents > App registrations > VIP Food App

VIP Food App | App roles

Search

+ Create app role | Got feedback?

App roles

App roles are custom roles to assign permissions to users or apps. The application defines as permissions during authorization.

How do I assign App roles

Display name	Description	Allowed member type
No app roles have been added.		

Display name *

EquipManagerRole

Allowed member types *

☒ Users/Groups

☐ Applications

☐ Both (Users/Groups + Applications)

Value *

EquipManagerRole

Description *

Tailored for the equipment manager, granting access to overseeing and managing equipment functionalities.

Do you want to enable this app role?

☒

Apply Cancel

5. Assign Application owners and confirm the selected owner(s).

Microsoft Azure | Search resources, services, and docs (G+)

Home > VIPEvents > App registrations > VIP Food App

VIP Food App | Owners

Search

+ Add owners | Remove owners | Got feedback?

The users listed here can view and edit this application registration. Additionally, any user (may not be listed here) with administrative privileges to manage any application (e.g., Global Administrator, Cloud App Administrator etc.) can view and edit the application registrations. Currently, only individual users are supported as owners of applications. Assignment of groups as owners is not yet supported. If the user setting "Restrict access to Microsoft Entra ID administration portal" is set to Yes, non-admin users will not be able to use the Azure portal to manage the applications they own. [Learn more](#)

No owners have been assigned to this application.

Add owners

Microsoft Azure | Search resources, services, and docs (G+)

Home > VIPEvents > App registrations > VIP Food App

VIP Food App | Owners

Search

Try changing or adding filters if you don't see what you're looking for.

Search

Search owner

All Users

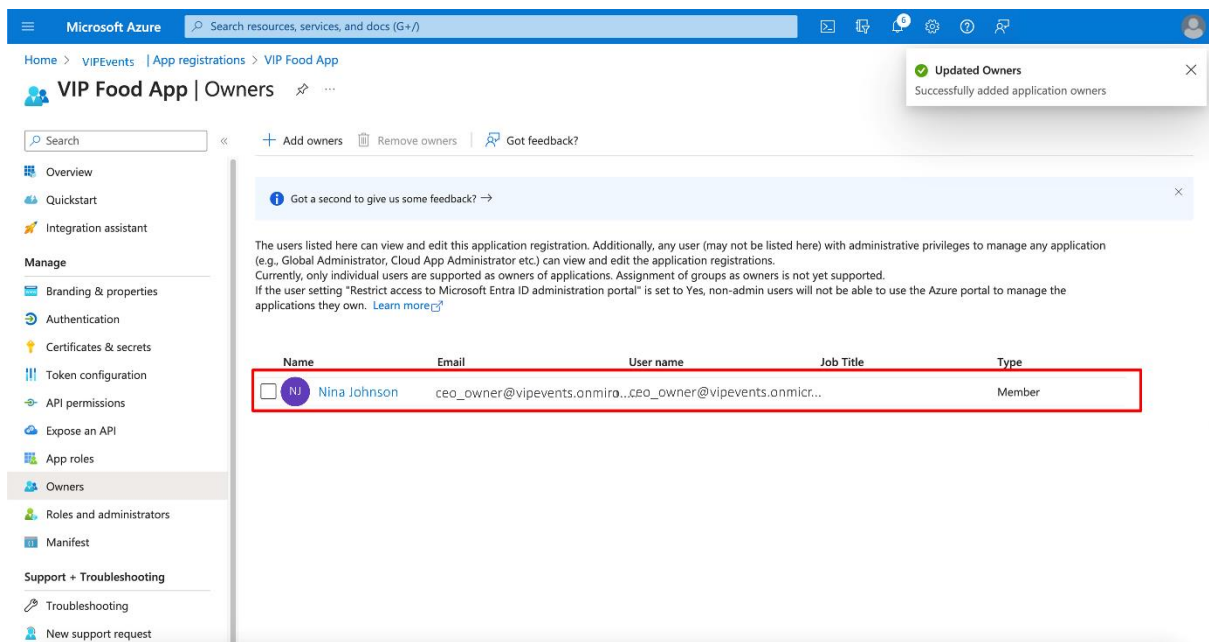
	Type	Details
☐	User	Lina Kim catering_manager@vipevents...
☐	User	Daniel Turner office_worker1@vipevents.on...
☐	User	Isabella Dube office_worker2@vipevents.on...
☐	User	Ava Miller office_worker3@vipevents.on...
☐	User	Worker 1 user1@vipevents.onmicrosoft...
☒	User	Nina Johnson ceo_owner@vipevents.onmicr...

Select

Selected (1)

Reset

Nina Johnson
ceo_owner@vipevents.onmicr...



The values associated with each role, are described below:

- Equipment handlers role:

Role Name: EquipHandlersRole

Details: Designed for equipment handlers, providing access to equipment management functionalities on the VIP Food app.

- Equipment manager role:

Role Name: EquipManagerRole

Details: Tailored for the equipment manager, granting access to overseeing and managing equipment functionalities.

- Chefsrole:

Role Name: ChefsRole

Details: Created for chefs, offering access to food preparation and kitchen management functionalities on the VIP Food app.

- Head chef role:

Role Name: HeadChefRole

Details: Specifically designed for the head chef, providing elevated access to overall kitchen management functionalities.

- Catering manager role:

Role Name: CateringManagerRole

Details: For catering managers, offering access to planning and managing catering events functionalities on the VIP Food app.

- Office workers role:

Role Name: OfficeWorkersRole

Details: Tailored for office workers, providing access to office-related activities and data management functionalities on the VIP Food app.

- CEO (Owner) role:

Role Name: CEORole

Details: Specifically designed for the CEO (owner), granting access to all functionalities on the VIP Food App for strategic oversight.

- Temporary role:

Role Name: TempRole

Details: This role will be assigned to temporary workers, and they will only have access to permission that the user needs for that day – it will be automatically disabled after a period.

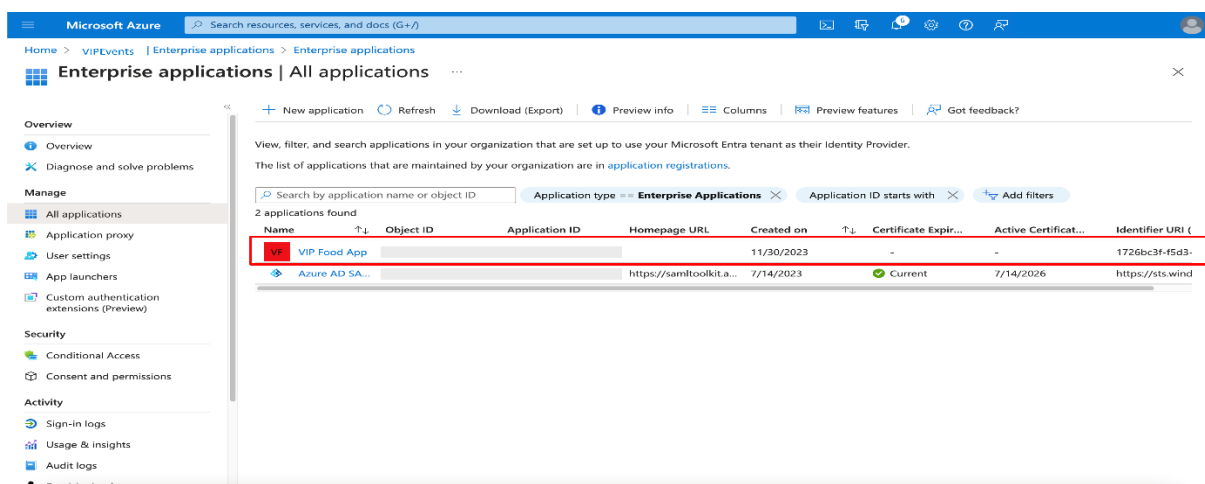
Assigning permissions to Azure roles(Task 3)

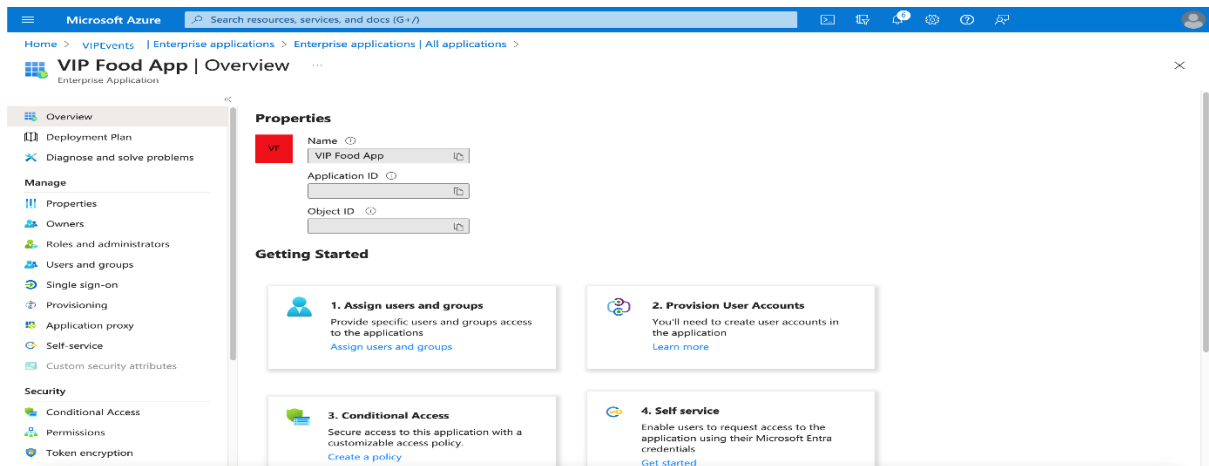
The details of the application and roles should be provided to application developers for the implementation of role-based access control. This step ensures seamless integration of roles into the application structure. They will assign the right role to the right controllers to assign the required permissions.

Mapping user groups to Azure roles (Task 4)

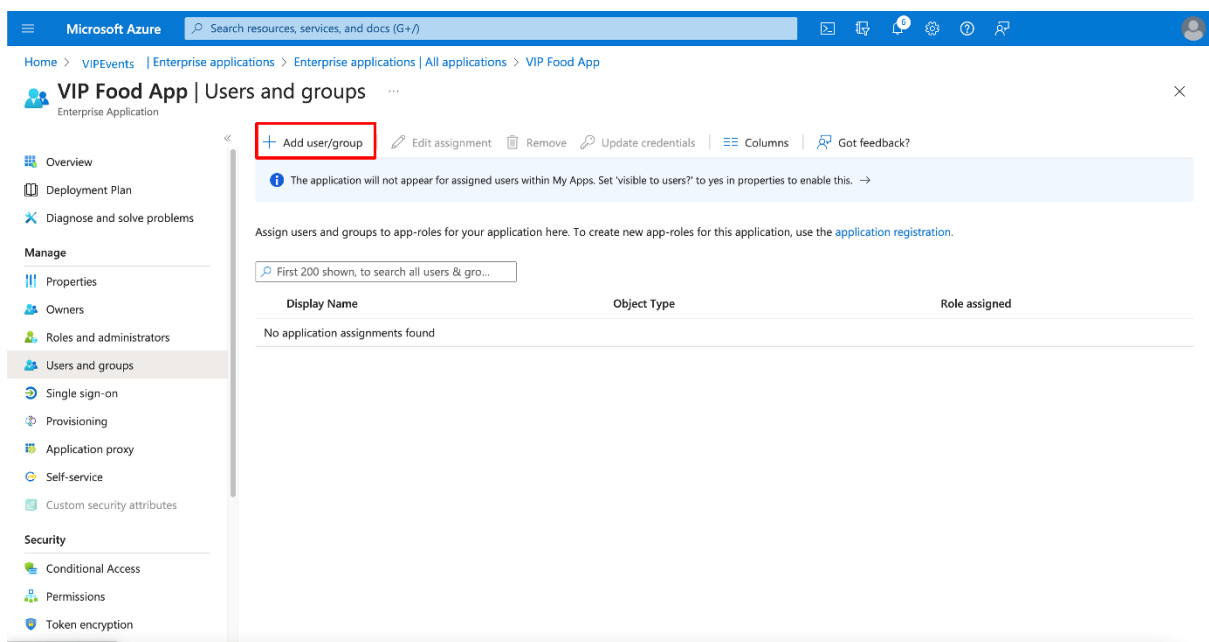
User groups should be mapped to Azure roles to ensure access permissions are tailored to specific roles and responsibilities within the organization. The steps to map user groups to Azure roles are outlined below.

1. Sign in to the **Microsoft Entra admin center**.
2. Go to **Manage**, select **Enterprise Applications** and then select the app.



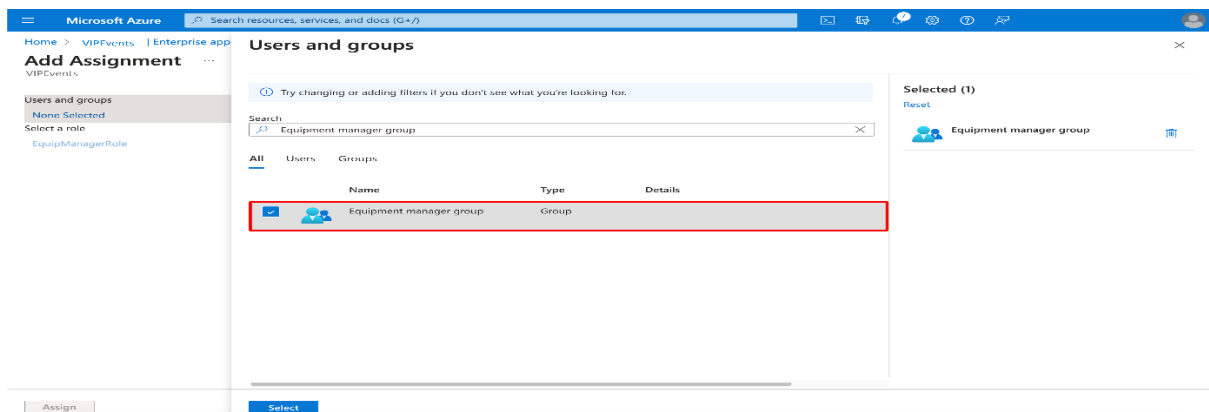


3. Add user group



4. Select Add assignments.

5. Select the group you want to assign to this role. Use the Role Mapper table provided after the screenshots to know which groups to add to which roles. Note that only role-assignable groups are included in the table. Select the role.



Microsoft Azure

Search resources, services, and docs (G+)

Home > VIPEvents > Enterprise applications > Enterprise applications | All applications > VIP Food App | Users and groups >

Add Assignment

...

VIPEvents

When you assign a group to an application, only users directly in the group will have access. The assignment does not cascade to nested groups.

Users and groups
1 group selected.
Select a role *
None Selected

Assign

Select a role

Only a single role can be selected

Enter role name to filter items...

EquipHandlersRole

EquipManagerRole

Selected Role
You haven't selected any role.

Select

Microsoft Azure

Search resources, services, and docs (G+)

Home > VIPEvents > Enterprise applications > Enterprise applications | All applications > VIP Food App | Users and groups >

Add Assignment

...

VIPEvents

When you assign a group to an application, only users directly in the group will have access. The assignment does not cascade to nested groups.

Users and groups
1 group selected.
Select a role *
EquipManagerRole

Assign

Microsoft Azure

Search resources, services, and docs (G+)

Home > VIPEvents > Enterprise applications > Enterprise applications | All applications > VIP Food App

VIP Food App | Users and groups

...

VIPEvents > Application

Application assignment succeeded
0 users & 1 group have been assigned access

+ Add user/group | Edit assignment | Remove | Update credentials | Columns | Got feedback?

The application will not appear for assigned users within My Apps. Set 'visible to users?' to yes in properties to enable this. →

Assign users and groups to app-roles for your application here. To create new app-roles for this application, use the [application registration](#).

First 200 shown, to search all users & gro...

	Display Name	Object Type	Role assigned
☐	Equipment manager group	Group	EquipManagerRole

Overview
Deployment Plan
Diagnose and solve problems
Manage
Properties
Owners
Roles and administrators
Users and groups
Single sign-on
Provisioning
Application proxy
Self-service
Custom security attributes
Security
Conditional Access
Permissions
Token encryption

6. If a group isn't listed, you will need to create a role-assignable group.

7. Select **Add** to assign the role to the group.

Role Mapper table:

Role	Group to be added
Equipment handler	Equipment handlersgroup
Equipment manager	Equipment manager group
Chef	Chefs group
Head chef	Head chef group
Catering manager	Catering manager group
Office worker	Office workers group
CEO	CEO (Owner) group
Tempworker (role name)	Worker - external

STAGE 04

AAD Connections (Stage 4)

This section provides specifications for testing Azure AD connections. By performing these tests, VIP Events can verify if new authentication and access procedures are implemented correctly.

Client app integration (Task 1)

- To test whether the Azure login functionality is integrated with the VIP Food app, log into the VIP Food app using a designated test user account.
- To test RBAC implementation, log into the VIP Food app with a user account that has been assigned the relevant Azure role through group membership.
- To verify if the Azure AD Authentication functionality is working, follow the Azure AD authentication prompts.
- Perform functional testing by entering Azure AD credentials and rigorously testing various functionalities of the app to ensure flawless operation.

User account validation (Task 2):

- Begin the Azure user account testing process by logging into the Azure AD portal using the designated credentials.
- Navigate to **User Overview** by going to **Users** and selecting **All users** for a comprehensive overview of user accounts.
- Scrutinize the user list to confirm the creation of all necessary accounts.
- To verify if the user attributes are correct, select a user account, choose **Edit**, and meticulously verify attributes like job title and department.

Application role assignment (Task 3)

- Navigate to **Enterprise Applications** and select **All applications** for an overview.
- Choose the **VIP Food App** and select **Roles and administrators**.
- Review the user-role assignments to ensure alignment with organizational needs.

Group/Role-based access testing (Task 4)

- To start the group-based access testing, create a test user account.
- Assign the test user to a specific user group.
- Test if the role is correctly assigned to the group and if the members of the respective group have the required role assigned automatically. To do this, attempt to access app features applicable to that particular group/role.

MFA verification (Task 5)

- Start by logging into the Azure AD portal.
- To navigate to MFA Settings, go to **Security** and select **Multi-Factor Authentication** for an overview.
- To review MFA configuration, scrutinize the user list to ensure MFA is enabled, and manage user settings for detailed MFA method verification.

Logging and monitoring (Task 7)

- To access Audit Logs navigate to **Monitoring** and select **Audit logs for comprehensive monitoring**.
- Scrutinize audit logs to identify irregular activities such as failed sign-ins or changes to user permissions.

Documentation review (Task 8)

- Meticulously review Azure AD configuration documentation.
- Verify that the setup in the Azure AD portal aligns with the documentation.
- Promptly update documentation to reflect any identified discrepancies.

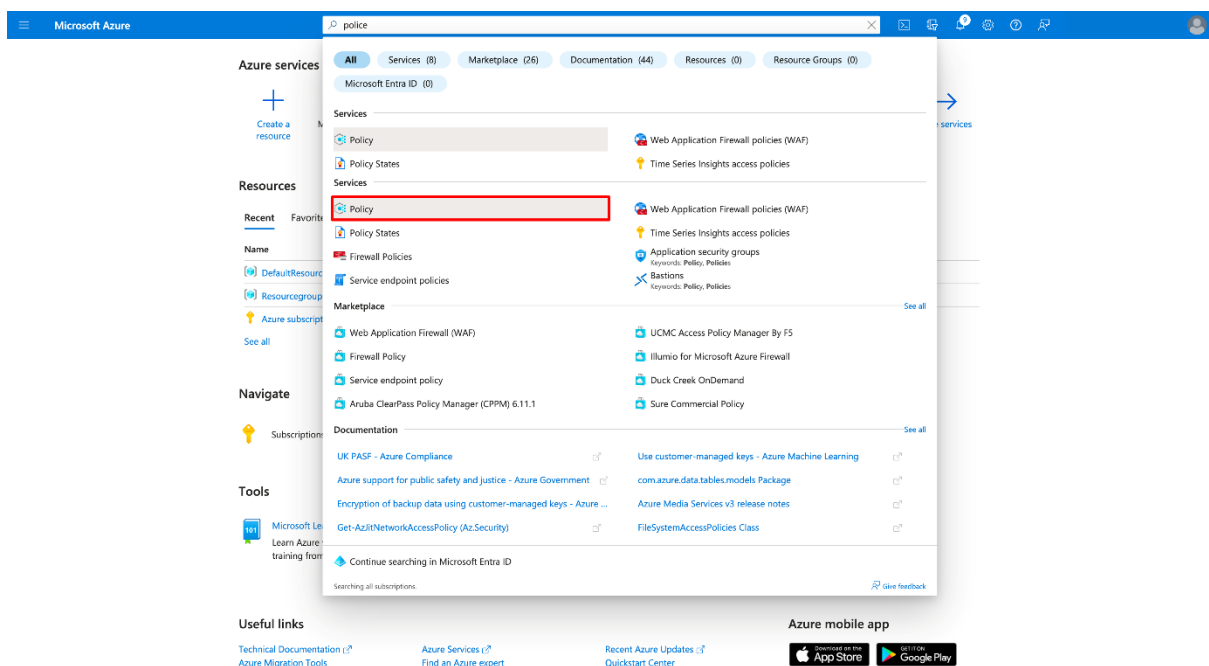
Policy implementation (Stage 5)

Through the creation and implementation of Azure Policies a robust security framework for VIP Events can be established. Specific policies for user authentication and network configuration for web applications need to be crafted. This section of the proposal provides step-by-step instructions for designing, implementing, and testing these policies to ensure a secure and compliant environment.

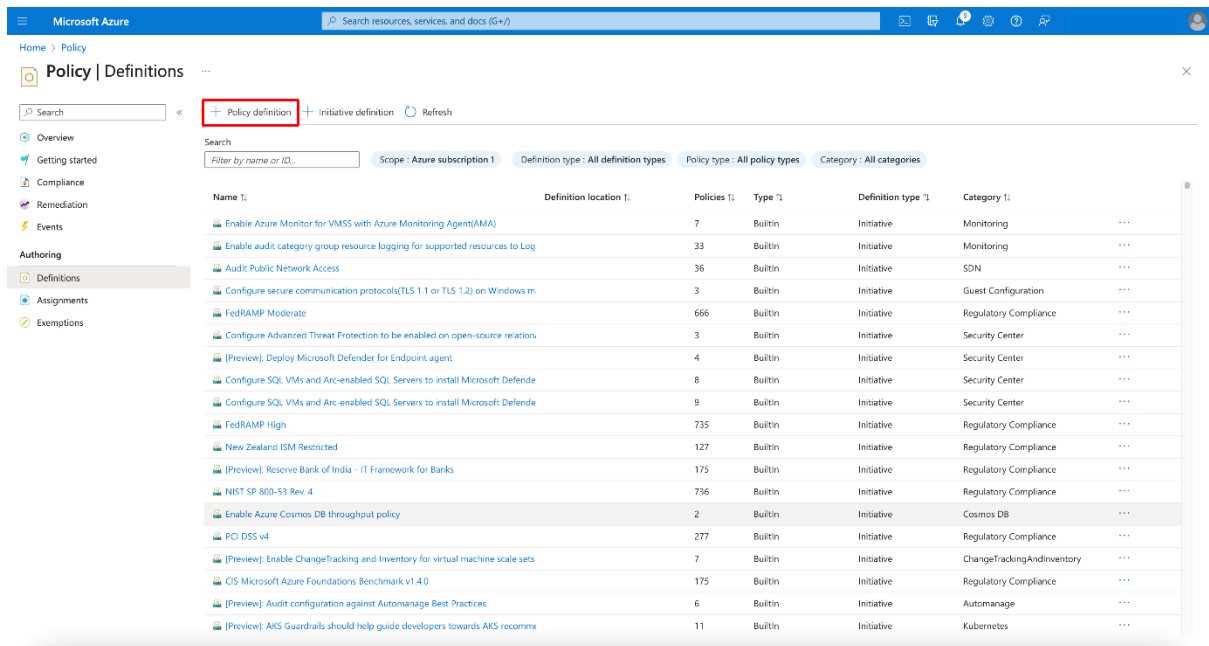
User authentication policy (Task 1)

These steps should be followed to create a user authentication policy.

1. Navigate to **Azure Policy** in the Azure Portal.
2. In the left-hand menu, select **All services**, and then search for and choose **Policy**.



3. To create a new policy definition, access the **Definitions** tab on the **Policy** page and select **+Policy definition**.



4. Fill in the necessary information like **Display Name**, **Description**, and **Category**.

BASICS

Definition location *

Name *

Description

Category ☒ Create new ☐ Use existing

Category

POLICY RULE

Import sample policy definition from GitHub

Learn more about policy definition structure

```

1 {
2   "mode": "All",
3   "policyRule": {
4     "if": {
5       "not": {
6         "field": "location",
7         "in": "[parameters('allowedLocations')]"
8       }
9     },
10    "then": {
11      "effect": "audit"
12    }
13  },

```

5. In the **Policy Rule** section, set parameters to enforce MFA for users.
6. Use the **Add a clause** option to specify conditions for the policy, such as requiring MFA for specific user roles or all users.
7. Review the policy settings and select **Review + create**.
8. Ensure validation passes, then select **Create** to create the policy definition.
9. Go to the **Assignments** tab and click on **+Assign policy** to assign the policy to a specific scope. Select the VIP Events subscription.

10. To specify assignment details provide details such as **Scope**, **Parameters**, and **Assignment name**.

11. Select **Review + assign** and then **Assign** to apply the policy.

Network configuration policy for web applications (Task 2)

These steps should be followed to configure network configurations for the VIP Food app.

1. Use the built-in policy named **App Service app slots should only be accessible over HTTPS**.
2. This policy can be accessed via this [link](#).

The screenshot displays the Microsoft Azure portal interface. At the top, the 'Microsoft Azure' header is visible with a search bar. Below the header, the page title is 'App Service app slots should only be accessible over HTTPS'. Underneath the title, there are tabs for 'Assign', 'Edit definition', 'Duplicate definition', and 'Delete definition'. The 'Essentials' section provides key details about the policy: Name, Description, Available Effects (Audit), Category (App Service), Definition location, Definition ID, Type (Built-in), and Mode (Indexed). Below this, the 'Definition' tab is active, showing the JSON policy rule. The JSON defines the policy's properties, metadata, parameters (effect, allowed values), and the policy rule itself, which targets function app slots.

```
1 {
2   "properties": {
3     "displayName": "App Service app slots should only be accessible over HTTPS",
4     "policyType": "BuiltIn",
5     "mode": "Indexed",
6     "description": "Use of HTTPS ensures server/service authentication and protects data in transit from network layer eavesdropping attacks.",
7     "metadata": {
8       "version": "2.0.0",
9       "category": "App Service"
10    },
11    "parameters": {
12      "effect": {
13        "type": "String",
14        "metadata": {
15          "displayName": "Effect",
16          "description": "Enable or disable the execution of the policy"
17        },
18        "allowedValues": [
19          "Audit",
20          "Disabled",
21          "Deny"
22        ],
23        "defaultValue": "Audit"
24      },
25    },
26    "policyRule": {
27      "if": {
28        "allOf": [
29          {
30            "field": "kind",
31            "notContains": "functionapp"
32          },
33          {
34            "field": "type",
35            "equals": "Microsoft.Web/sites/slots"
36          },
37          {
38            "anyOf": [
```

3. Select **Assign**, then choose **Scope** and opt for VIP Food App Subscription.

The screenshot shows the Microsoft Azure portal interface for assigning a policy. The breadcrumb trail is 'Home > App Service app slots should only be accessible over HTTPS >'. The main heading is 'App Service app slots should only be accessible over HTTPS' with a sub-label 'Assign policy'. Below this are tabs for 'Basics', 'Advanced', 'Parameters', 'Remediation', 'Non-compliance messages', and 'Review + create'. The 'Basics' tab is active, showing fields for 'Scope' (with a link 'Learn more about setting the scope'), 'Exclusions' (with a link 'Optionally select resources to exclude from the policy assignment'), 'Policy definition' (set to 'App Service app slots should only be accessible over HTTPS'), 'Assignment name' (set to 'App Service app slots should only be accessible over HTTPS'), 'Description' (empty text area), 'Policy enforcement' (set to 'Enabled'), and 'Assigned by' (set to 'Jamie'). At the bottom, the 'Review + create' button is highlighted with a red box, along with 'Cancel', 'Previous', and 'Next' buttons.

1. Click Select **Review+Create** to assign the same

Testing in a non-production environment (Task 3)

Before applying policies to production resources, testing in a non-production environment is crucial. Testing identifies unintended consequences or disruptions, verifies policy effectiveness, and ensures alignment with organizational requirements.

Implementing policies within a system carries inherent risks that can compromise its functionality and security. There is a possibility that policies may inadvertently block access for legitimate users or resources, leading to disruptions in normal operations. Unforeseen issues in policy configuration may arise, causing service disruptions and potentially compromising the integrity of the system. Testing policies directly in a production environment can pose security and compliance risks.

The benefits of testing policies before full implementation are substantial. Early detection of misconfigurations or conflicts in policies allows for timely adjustments, ensuring a smoother implementation process. Testing also verifies the impact of policies on resources and user access, helping to identify potential issues before they escalate. By testing policies before implementation, organizations can minimize disruptions to production services, promoting a more stable and secure operational environment.

Through comprehensive policies and thorough testing in a non-production environment, the organization can mitigate security risks, protect sensitive data, and ensure the integrity of its IT systems and operations. Regular monitoring and updates to policies will further enhance the security posture over time.

Conclusion

This is a summary of the cybersecurity proposal for VIP Events:

In the company requirements section of the proposal, crucial information about VIP Events' business operations, security requirements, and user roles was summarized, laying the groundwork for the subsequent stages. The proposal included the creation of an Azure AD tenant for VIP Events. It also detailed how user accounts were to be configured and group-based access control implemented, in line with the specific security requirements identified.

Further, the proposal outlined roles and access configuration. It specified how access permissions for each user group should be finely adjusted to ensure precision, guaranteeing that users have the appropriate level of access for the seamless execution of their job responsibilities.

The proposal also covered the integration of Azure AD with key business applications, particularly the VIP Food app. This integration was designed to streamline user authentication and authorization, fostering a cohesive and efficient operational environment.

Lastly, the proposal recommended implementing certain policy configurations to help VIP Events mitigate security risks, protect sensitive data, and ensure the integrity of its IT systems and operations. The policies described are intended to govern user behavior, regulate data access, and manage devices, thereby forming a robust framework to safeguard VIP Events' cybersecurity landscape.

In summary, the proposal outlines a structured and comprehensive approach to strengthen VIP Events' security framework, safeguarding against potential threats and ensuring ongoing compliance with industry standards.